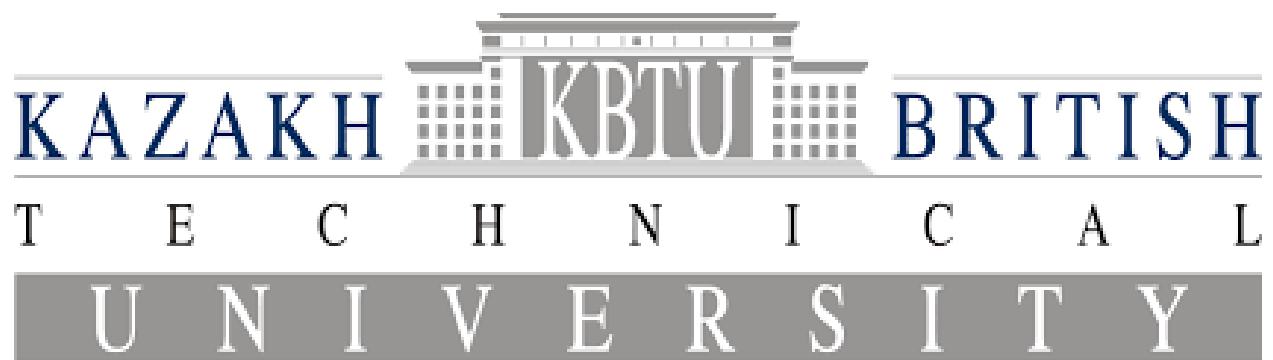




School of Information Technology and Engineering
Security of Operating System

Deploying Wazuh
“Secure SWIFT Logical Terminal Infrastructure”

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Almaty, 2026

Topic

Target

The target of this task is to deploy a centralized logging and monitoring solution that allows me to collect and observe all system logs and metrics in one place. For this purpose, I will use Wazuh as the main platform.

My goal is to configure several machines with monitoring agents and connect them to a central server, where all logs, system activities, and performance metrics are aggregated. The collected data will be processed and stored using OpenSearch.

The single point of observation for the entire system will be the Wazuh Dashboard, where I will be able to monitor events in real time. Through this dashboard, I will track system logs, analyze security alerts, and observe key performance metrics such as CPU, memory, and disk usage.

By completing this setup, I aim to have a clear and centralized view of my infrastructure, enabling me to quickly detect issues, monitor system behavior, and respond to potential security incidents efficiently.

Step 1: Installation and Deployment of Wazuh

In the first step, I deployed the centralized monitoring platform using Wazuh on a virtual machine. The installation was performed on a Linux-based system, where I set up all core components required for the system to operate.

The deployment included the installation and configuration of the following components:

- Wazuh Manager – responsible for collecting and analyzing data from agents
- OpenSearch – used for indexing and storing logs
- Wazuh Dashboard – web interface for visualization and monitoring

All services were successfully started and verified to be running on the virtual machine. After installation, I accessed the Wazuh Dashboard through a web browser and confirmed that the system was operational.

Oracle VirtualBox Manager

FileMachineHelp

NewOpenSettingsDiscardShow

swift-terminal1

Running

swift-support1

Running

swift-terminal1 Clone

Powered Off

Wazuh OVA

Running

Details

General

System

Display

Storage

Audio

Network

USB

Shared folders

Description

Base Memory: 8192 MB

Processors: 4

Boot Order: Floppy, Optical, Hard Disk

Acceleration: Nested Paging, KVM Paravirtualization

Video Memory: 16 MB

Graphics Controller: VBoxVGA

Remote Desktop Server: Disabled

Recording: Disabled

Controller: IDE

IDE Secondary Device 0: wazuh-4.14.4-disk-1.vdi (Normal, 25.00 GB)

Controller: Floppy

Floppy Device 0: Empty

Host Driver: Default

Controller: ICH AC97

Adapter 1: Intel PRO/1000 MT Server (Bridged Adapter, MediaTek Wi-Fi 6 MT7921 Wireless LAN Card)

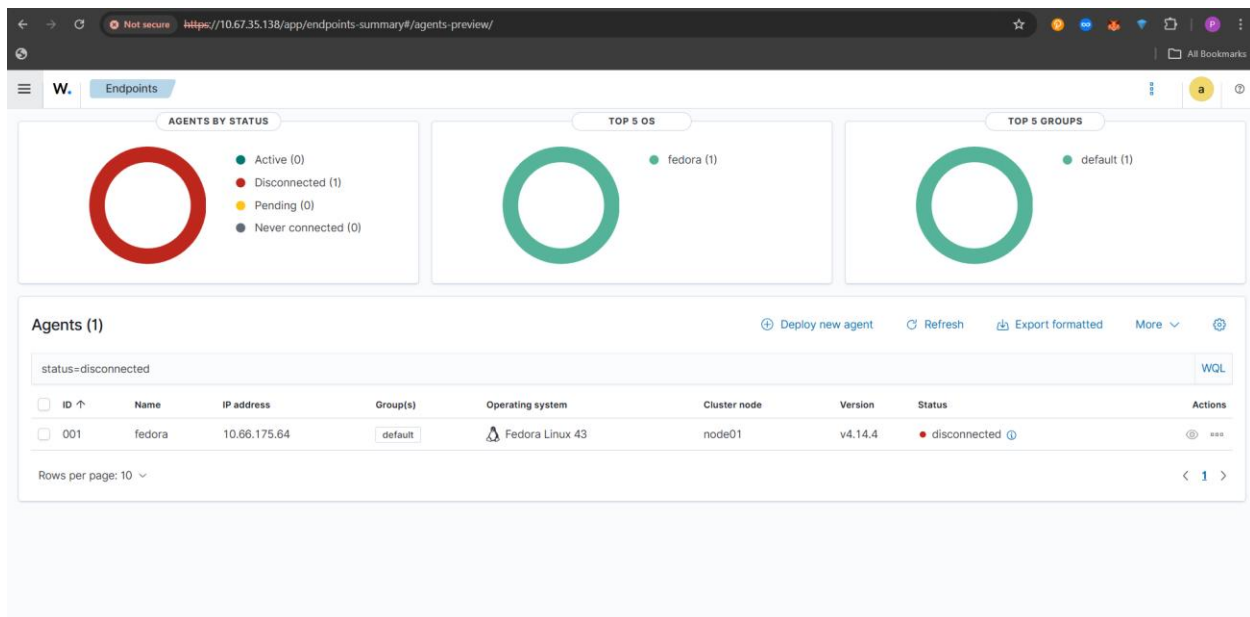
Disabled

None

Wazuh enhances security visibility in your infrastructure by monitoring endpoints at the operating system and application file integrity monitoring, intrusion detection, and compliance monitoring.

```
Wazuh OVA [Running] - Oracle VirtualBox
File Machine View Input Devices Help

link/ether 08:00:27:b6:b7:ee brd ff:ff:ff:ff:ff:ff
altname enp0s17
inet 10.67.35.138/24 metric 1024 brd 10.67.35.255 scope global dynamic eth0
    valid_lft 3599sec preferred_lft 3599sec
inet6 fe80::a00:27ff:feb6:b7ee/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
[root@wazuh-server ~]# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b6:b7:ee brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.67.35.138/24 metric 1024 brd 10.67.35.255 scope global dynamic eth0
        valid_lft 3557sec preferred_lft 3557sec
    inet6 fe80::a00:27ff:feb6:b7ee/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[root@wazuh-server ~]# ip aa
Object "aa" is unknown, try "ip help".
[root@wazuh-server ~]# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b6:b7:ee brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.67.35.138/24 metric 1024 brd 10.67.35.255 scope global dynamic eth0
        valid_lft 3553sec preferred_lft 3553sec
    inet6 fe80::a00:27ff:feb6:b7ee/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[root@wazuh-server ~]#
```



Step 2: Installation and Configuration of Wazuh Agents

In the second step, I configured additional virtual machines to send their logs and system metrics to the centralized monitoring server. For this purpose, I installed the Wazuh agent on two separate virtual machines.

Swift-terminal:

```
^Cswift-terminal@fedora:/opt/swift-backend$ sudo dnf install wazuh-agent -y
Updating and loading repositories:
  Fedora 43 - x86_64 - Updates          100% | 1.4 KiB/s | 6.1 KiB | 00m04s
Repositories loaded.
Package "wazuh-agent-4.14.4-1.x86_64" is already installed.

Nothing to do.
swift-terminal@fedora:/opt/swift-backend$
```

Swift-support:

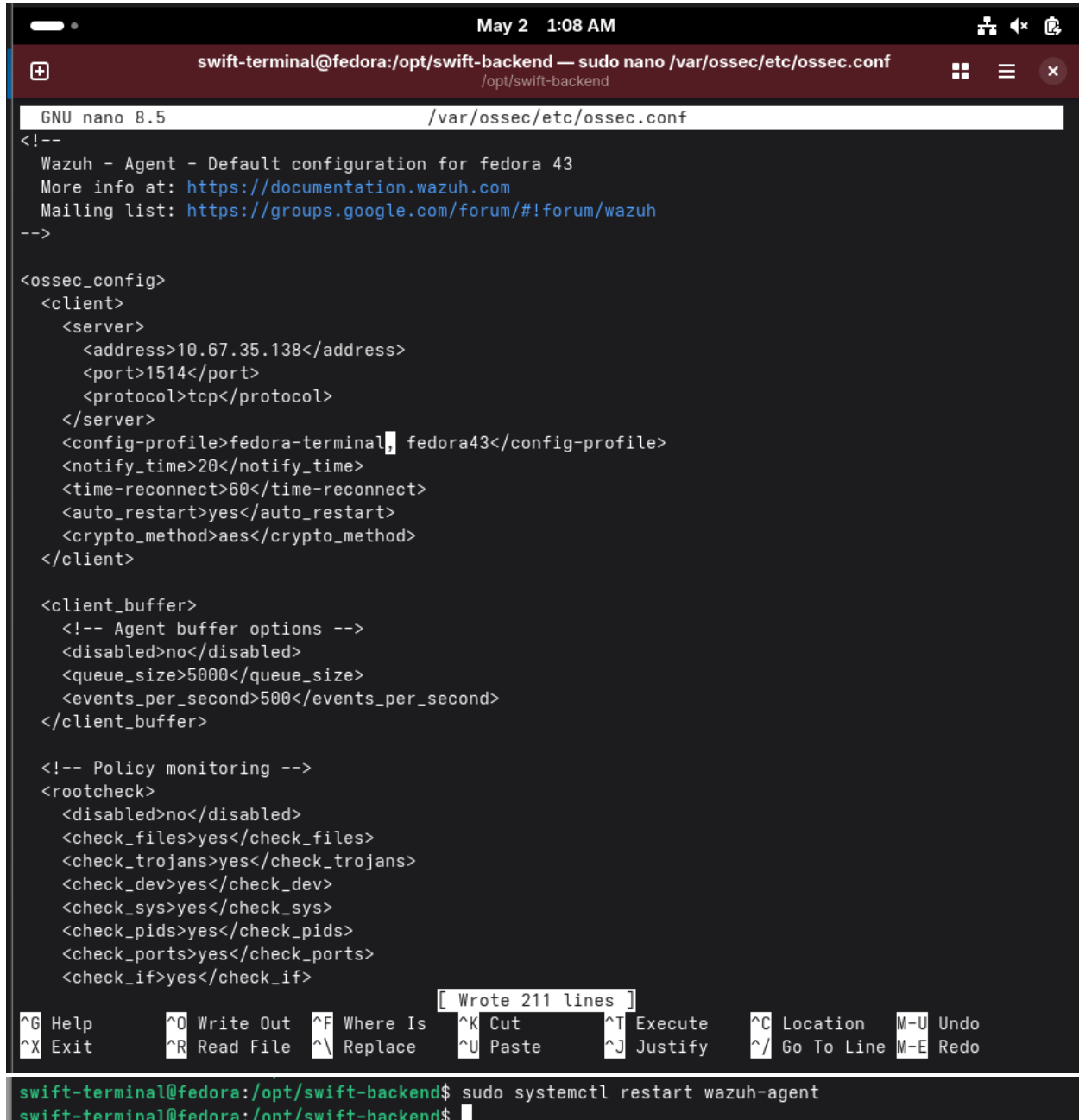
```
^Cswift-support@swift-support:/$ sudo dnf install wazuh-agent -y
Updating and loading repositories:
  Fedora 43 - x86_64 - Updates          100% | 1.4 KiB/s | 6.1 KiB | 00m04s
Repositories loaded.
Package "wazuh-agent-4.14.4-1.x86_64" is already installed.

Nothing to do.
```

After installing the agent on each machine, I configured them to connect to the central Wazuh server by specifying the IP address of the Wazuh Manager. This

allows the agents to securely forward collected data, including system logs and performance metrics, to the central system.

Swift-terminal:



The screenshot shows a terminal window titled "swift-terminal@fedora:/opt/swift-backend — sudo nano /var/ossec/etc/ossec.conf". The window displays the configuration for the Wazuh agent in a nano editor. The configuration includes comments about Wazuh, the agent's role, and links to documentation and mailing lists. It defines the `<ossec_config>` section with client settings like server address, port, protocol, config profile, notify time, reconnect time, auto restart, and crypto method. It also defines the `<client_buffer>` section with options for disabled status, queue size, and events per second. Finally, it defines the `<rootcheck>` section with options for disabled status and various checks like files, trojans, dev, sys, pids, ports, and if. A status bar at the bottom indicates "Wrote 211 lines". Below the nano editor, the terminal shows the command `sudo systemctl restart wazuh-agent` being executed.

```
GNU nano 8.5 /var/ossec/etc/ossec.conf
<!--
Wazuh - Agent - Default configuration for fedora 43
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <client>
    <server>
      <address>10.67.35.138</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>fedora-terminal fedora43</config-profile>
    <notify_time>20</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
  </client>

  <client_buffer>
    <!-- Agent buffer options -->
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Policy monitoring -->
  <rootcheck>
    <disabled>no</disabled>
    <check_files>yes</check_files>
    <check_trojans>yes</check_trojans>
    <check_dev>yes</check_dev>
    <check_sys>yes</check_sys>
    <check_pids>yes</check_pids>
    <check_ports>yes</check_ports>
    <check_if>yes</check_if>

[ Wrote 211 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo

swift-terminal@fedora:/opt/swift-backend$ sudo systemctl restart wazuh-agent
swift-terminal@fedora:/opt/swift-backend$
```

Swift-support:

```
May 2 1:18 AM
swift-support@swift-support:/ — sudo nano /var/ossec/etc/ossec.conf
GNU nano 8.5 /var/ossec/etc/ossec.conf Modified
<!--
Wazuh - Agent - Default configuration for fedora 43
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

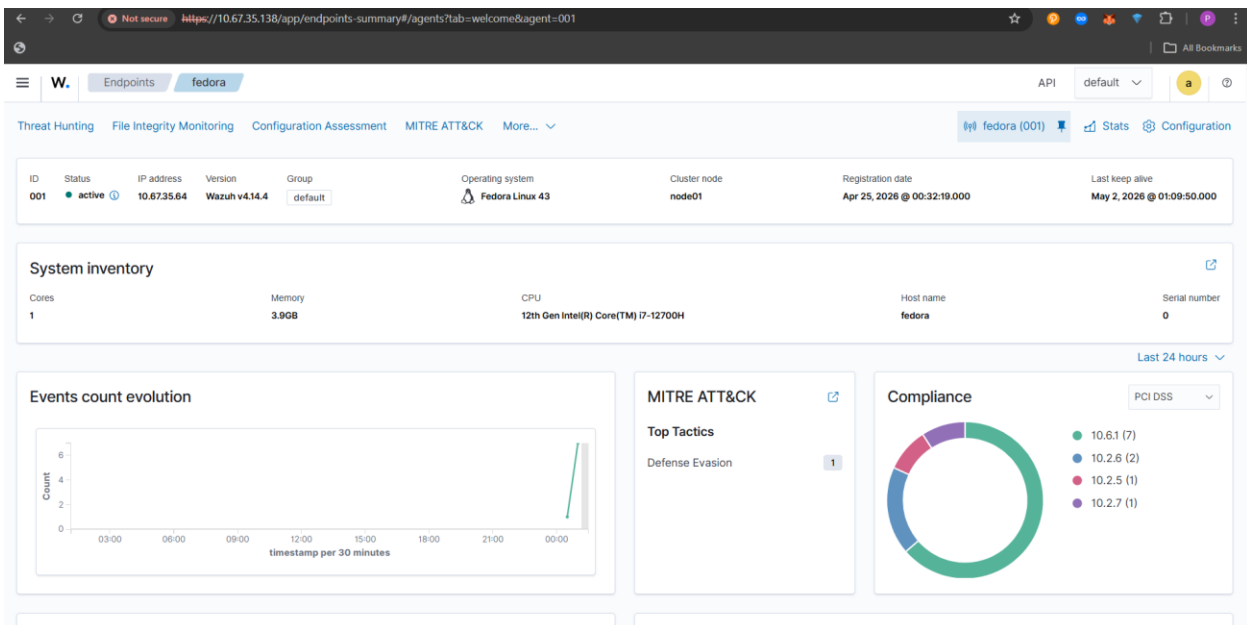
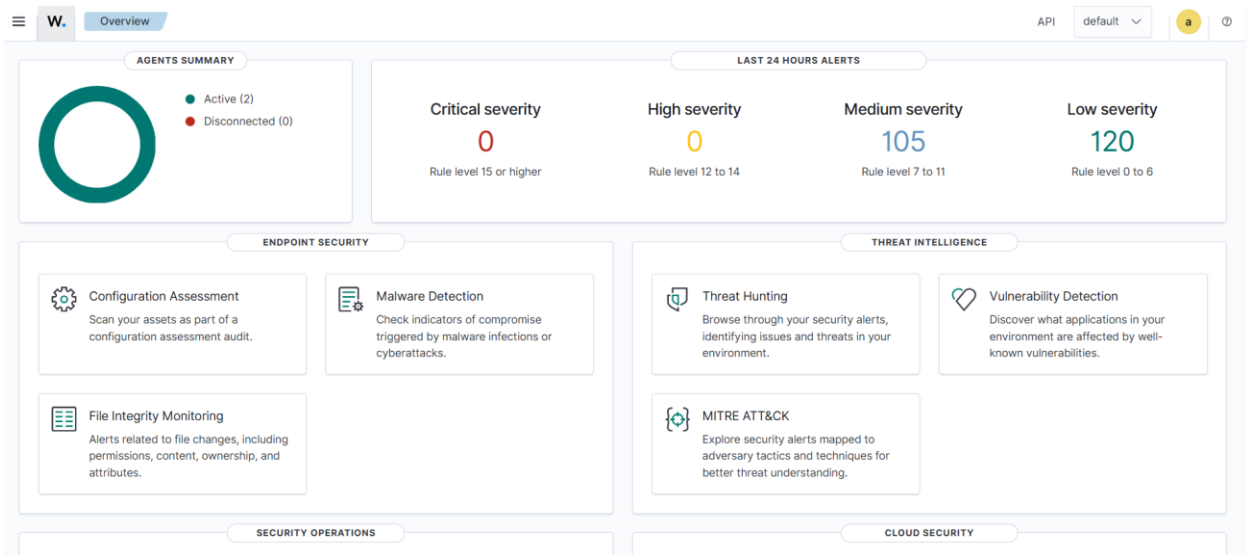
<ossec_config>
  <client>
    <server>
      <address>10.67.35.138</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>fedora, fedora43</config-profile>
    <notify_time>20</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
  </client>

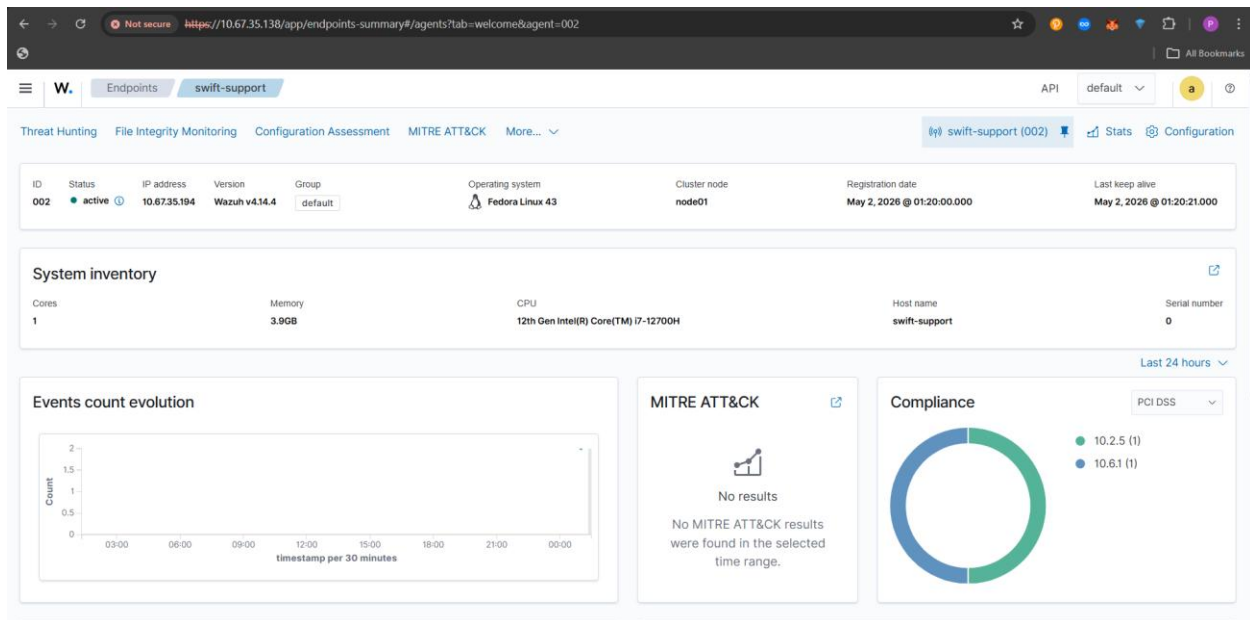
  <client_buffer>
    <!-- Agent buffer options -->
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>

^G Help      ^O Write Out  ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify

swift-support@swift-support:/$ sudo systemctl restart wazuh-agent
```

Once configured, I started the agent services on both virtual machines and verified their connectivity. The agents successfully registered with the Wazuh Manager and became visible in the Wazuh Dashboard.





This step ensures that multiple systems are now integrated into the centralized monitoring environment, enabling real-time data collection and unified visibility across all connected machines.

Conclusion

In this work, I successfully deployed a centralized monitoring and logging system using Wazuh. The system was designed to collect, process, and visualize logs and metrics from multiple virtual machines within a unified environment.

During the implementation, I installed and configured the Wazuh server components, including OpenSearch and the Wazuh Dashboard, which serves as the central interface for monitoring. I then connected two additional virtual machines by installing Wazuh agents and configuring them to communicate with the central server.

As a result, all connected systems are now able to send logs and performance data to a single platform, where they can be analyzed in real time. The dashboard provides a clear and structured view of system activity, allowing me to monitor events, detect anomalies, and respond to potential security threats efficiently.

This deployment demonstrates the effectiveness of centralized monitoring in improving system visibility and security. It also provides a scalable foundation that can be extended by adding more agents, defining custom alerting rules, and integrating additional data sources in future work.